

DSM-Project

Topic: Self Balancing and Walking Humanoid Robot Simulation which includes - ~~the~~ Lower Body Joint Analysis, Torque analysis.

1. Modeling Scope:

- lower Body rigid-Body model with ^{two} ~~two~~ legs two 6-DOF legs attached to a Torso link.
- Euler-lagrange formulation is written in generalized co-ordinate form.

2. Core Physical Parameters:

Parameters	Values
Total Robot mass	5.95 kg
Gravity	9.8 m/s ²
Body mass	4.54 kg
R-leg mass	0.704 kg
L-leg mass	0.704 kg
Thigh length L1	0.18 m
Shin length L2	0.18 m
Right Hip offset	[0, -0.065, -0.175]
left hip offset	[0, 0.065, -0.175]
Controller sample time	1/240 sec

3- Generalized Co-ordinates:

Coordinate	Joint	Axis	Joint Origin (cm)
q_1	Hip Yaw	z	$[0, -0.065, -0.175]$
q_2	Hip Roll	x	$[0, 0, 0]$
q_3	Hip Pitch	y	$[0, 0, 0]$
q_4	Knee Pitch	y	$[0, 0, -0.18]$
q_5	Ankle Pitch	y	$[0, 0, -0.18]$
q_6	Ankle Roll	x	$[0, 0, 0]$

→ For one leg the generalized coordinate vector is obtained as:

$$q_L = [q_1, q_2, q_3, q_4, q_5, q_6]^T$$

q_L = coordinates
for Right leg.

→ For 2 leg coordinates are given as:

$$q = [q_L^T, q_R^T]^T$$

→ the Range lies between
for All coordinates

-2.5 to 2.5.

4. ~~the~~ Euler

4. Euler-lagrange Kinematic Ingredients:

→ The right leg forward chain is obtained by:

• forward ^{transform} chain : $T_{06} = T_{01} T_{12} T_{23} T_{34} T_{45} T_{56}$

• Rodrigues formula used for each revolute Joint Rotation

$$R_{(C_i, \theta)} = I + [a] \sin \theta + [a]^2 (1 - \cos \theta)$$

• Link center of mass position and Velocity Definitions:

$$P_{C_i} = P_i + R_i C_i ; \quad V_{C_i} = J_{V_i}(q) \dot{q} ; \quad \omega_i = J_{\omega_i}(q) \dot{q}$$

5. Kinetic and Potential Energy:

→ single link kinetic Energy:

$$T_i = \frac{1}{2} m_i V_{C_i}^T V_{C_i} + \frac{1}{2} \omega_i^T I_{C_i} \omega_i$$

→ Total Kinetic energy: [Only 1 leg] $\Rightarrow T = \sum_{i=1}^6 T_i$

→ Gravitational Potential Energy: $V = \sum_{i=1}^6 m_i g z_{C_i}$

→ lagrangian : $L = T - V$

6. Euler-lagrange Equations of motion:

→ General Euler-lagrange Equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i$$

→ Generalized force vector with Actuator:

$$Q = \tau + J_c^T \lambda - D\dot{q} - \tau_f$$

→ Manipulator form: $M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + J_c^T \lambda$

7. Contact and Ground Reaction:

→ Contact contribution to Generalize Forces

$$Q_c^T = J_c^T \lambda$$

→ Point-Contact Interpretation at the foot:

$$\lambda = [N \ F_x \ F_y]^T$$

8. Parameter mapping for one Right leg:

Link	mass	com(m)	Principle Inertial terms.
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→ r-yaw-hip	0.108	[0, 0, 0]	$I_{xx} = 0.00004, I_{yy} = 0.00004$ $I_{zz} = 0.00005$
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→ r-roll-hip	0.108	[0, 0, 0]	$I_{xx} = 0.00005, I_{yy} = 0.00004$ $I_{zz} = 0.00004$
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→ r-thigh	0.128	[0, 0, -0.09]	$I_{xx} = 0.0005, I_{yy} = 0.0005$ $I_{zz} = 0.00002$
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→ r-knee	0.123	[0, 0, -0.09]	$I_{xx} = 0.0003, I_{yy} = 0.0003$ $I_{zz} = 0.00002$
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r-ankle	0.09	0	
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→ r-ankle	0.09	[0, 0, 0.01]	$I_{xx} = 0.00002, I_{yy} = 0.00002$ $I_{zz} = 0.00001$
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→ r-tip	0.139	[0.002, 0, -0.013]	$I_{xx} = 0.00008, I_{yy} = 0.0001$ $I_{zz} = 0.0002$
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